MTH636 – Game Theory

Lecture 32: Agent Form Games, Correlated Equilibrium & Auction Theory

1 Agent Form Representation & Bayesian Nash Equilibrium

In Lecture 31, we introduced Harsanyi games of incomplete information and defined the concept of **Bayesian Nash Equilibrium (BNE)**. We now establish the formal mathematical connection between a Harsanyi game and a strategic form game called the **Agent Form Game**, which will allow us to prove both the existence of BNE and its fundamental equivalence with Nash Equilibrium.

1.1 Formulation of the Agent Form Game

Let $\Gamma = \langle N, (T_i)_{i\in N}, p, S, (u_t)_{t\in T}\rangle$ be an incomplete information game, where:

- $N = \{1, 2, \ldots, n\}$ is the set of players.
- T_i is the finite type set of player i, and $T = \prod_{i=1}^{n} T_i$.
- p is the common prior probability distribution over T with $p(t) > 0$ for all $t \in T$.
- $A_i(t_i)$ is the action set available to player i when their type is t_i.
- $u_i(t, a)$ is player i's payoff given type profile $t \in T$ and action profile $a \in \prod_i A_i(t_i)$.

We formulate an associated strategic form game called the **Agent Form Game**, denoted by $\hat{\Gamma}$.

Definition: Agent Form Game $\hat{\Gamma}$

The **Agent Form Game** associated with the incomplete information game Γ is defined as the tuple:

$$\hat{\Gamma} = \langle \hat{N}, (\hat{S}_\ell)_{\ell\in\hat{N}}, (\hat{u}_\ell)_{\ell\in\hat{N}}\rangle$$

where:

1. **Set of Agents:** Each type of each player is treated as an autonomous decision-maker:
$$\hat{N} = \bigcup_{i=1}^{n} T_i$$
Thus, an agent $\ell \in \hat{N}$ corresponds to a specific pair (i, t_i) where $t_i \in T_i$.
2. **Strategy Sets:** If agent ℓ corresponds to type $t_i \in T_i$ of player i, their pure strategy set is simply the action set of that type:
$$\hat{S}_\ell = A_i(t_i) \quad \text{where } \ell = t_i$$
3. **Payoff Functions:** When an action profile $a = (a_\ell)_{\ell\in\hat{N}}$ is chosen by all agents, the payoff of agent $\ell = t_i \in T_i$ is the **interim expected utility** of player i

conditional on having type t_i:

$$\hat{u}_\ell(a) = \sum_{t_{-i} \in T_{-i}} p(t_{-i} \mid t_i)\, u_i\Big((t_i, t_{-i}),\, (a_{t_i}, (a_{t_j})_{j \neq i})\Big)$$

where $p(t_{-i} \mid t_i) = \frac{p(t_i, t_{-i})}{\sum_{t'_{-i}} p(t_i, t'_{-i})}$ is the conditional probability of opponent types.

Note

Remark on Strategy Correspondences: For any player $i \in N$, a behavior strategy σ_i specifies an action distribution $\sigma_i(t_i) \in \Delta(A_i(t_i))$ for each type $t_i \in T_i$. In the agent form game $\hat{\Gamma}$, each such distribution is precisely a mixed strategy for agent $\ell = t_i$. Conversely, any profile of mixed strategies for the agents $\{t_i\}_{t_i \in T_i}$ in the agent form game directly defines a valid behavior strategy for player i in the incomplete information game Γ.

1.2 Equivalence and Existence Theorems

Theorem: Equivalence of BNE and Agent Form NE

A behavior strategy profile $(\sigma_i)_{i \in N}$ is a **Bayesian Nash Equilibrium (BNE)** in the incomplete information game Γ if and only if the corresponding strategy vector $(\sigma_i(t_i))_{i \in N, t_i \in T_i}$ is a **Nash Equilibrium** in the agent form game $\hat{\Gamma}$.

Proof. ($\Longrightarrow$) Suppose $(\sigma_i)_{i \in N}$ is a BNE of Γ. By definition, for every player $i \in N$ and every type $t_i \in T_i$:

$$U_i(\sigma \mid t_i) \geq U_i((a_i, \sigma_{-i}) \mid t_i) \quad \forall\, a_i \in A_i(t_i)$$

Since the payoff function of agent $\ell = t_i$ in $\hat{\Gamma}$ is defined identically as the interim expected utility $\hat{u}_\ell = U_i(\cdot \mid t_i)$, no agent ℓ can profitably deviate from $\sigma_i(t_i)$. Thus, the strategy vector is a Nash Equilibrium of $\hat{\Gamma}$.

($\Longleftarrow$) We prove the contrapositive. Suppose $(\sigma_i)_{i \in N}$ is **not** a BNE of Γ. Then there exists some player $i \in N$, some type $t_i \in T_i$, and an action $a \in A_i(t_i)$ such that deviating to a yields a strictly higher interim utility:

$$U_i(\sigma \mid t_i) < U_i((a, \sigma_{-i}) \mid t_i)$$

In the agent form game $\hat{\Gamma}$, this implies that agent $\ell = t_i$ can unilaterally deviate to action a and achieve:

$$\hat{u}_\ell\Big((\sigma_j(t_j))_{j \in N, t_j \in T_j}\Big) < \hat{u}_\ell(a,\, \sigma_{-\ell})$$

Hence, the strategy profile is not a Nash Equilibrium of $\hat{\Gamma}$. □

Theorem: Existence of Bayesian Nash Equilibrium

Every finite incomplete information game Γ possesses at least one **Bayesian Nash Equilibrium** in behavior strategies.

Proof. The agent form game $\hat{\Gamma}$ is a standard finite strategic form game with player set $\hat{N}$ and pure action sets $\hat{S}_\ell$. By Nash's Existence Theorem (1950), every finite normal form game has at least one Nash Equilibrium in mixed strategies. By the Equivalence Theorem above, this mixed Nash Equilibrium in $\hat{\Gamma}$ is precisely a Bayesian Nash Equilibrium in behavior strategies for Γ. □

Theorem: BNE and Strategic Form Nash Equilibrium

In any Harsanyi incomplete information model with strictly positive common prior $p(t) > 0$ for all $t \in T$, a strategy profile is an ex-ante Nash Equilibrium if and only if it is a Bayesian Nash Equilibrium.

1.3 Worked Example: Constructing the Agent Form Game

Example: Agent Form Representation with Asymmetric Types

Consider an incomplete information game with two players, $N = \{1, 2\}$:

- Player 1 has two types: $T_1 = \{t_1, t_2\}$ (often denoted l_1, l_2).
- Player 2 has one type: $T_2 = \{\text{II}\}$.
- Common prior probabilities: $p(t_1, \text{II}) = \frac{1}{2}$ and $p(t_2, \text{II}) = \frac{1}{2}$.

Conditional beliefs are trivially:

$$p(\text{II} \mid t_1) = 1, \quad p(\text{II} \mid t_2) = 1, \quad p(t_1 \mid \text{II}) = \frac{1/2}{1} = \frac{1}{2}, \quad p(t_2 \mid \text{II}) = \frac{1}{2}$$

The state payoff matrices are:

State (t_1, II)

	L	R
T_1	$(1, 0)$	$(0, 2)$
B_1	$(0, 3)$	$(1, 0)$

State (t_2, II)

	L	R
T_2	$(0, 2)$	$(1, 1)$
B_2	$(1, 0)$	$(0, 2)$

Step 1: Formulate the Agent Form Game $\hat{\Gamma}$:

- Set of agents: $\hat{N} = \{t_1, t_2, \text{II}\}$ (3 agents).
- Action sets: $\hat{S}_{t_1} = \{T_1, B_1\}$, $\hat{S}_{t_2} = \{T_2, B_2\}$, $\hat{S}_{\text{II}} = \{L, R\}$.

Step 2: Payoff Computations for Profiles: Consider the pure action profile (T_1, B_2, L):

- For agent t_1:
$$\hat{u}_{t_1}(T_1, B_2, L) = 1 \cdot u_1((t_1, \text{II}), (T_1, L)) = 1$$
- For agent t_2:
$$\hat{u}_{t_2}(T_1, B_2, L) = 1 \cdot u_1((t_2, \text{II}), (B_2, L)) = 1$$

- For agent II:

$$\begin{aligned}\hat{u}_{\text{II}}(T_1, B_2, L) &= u_2((t_1, \text{II}), (T_1, L))\, p(t_1 \mid \text{II}) + u_2((t_2, \text{II}), (B_2, L))\, p(t_2 \mid \text{II}) \\ &= 0 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = 0\end{aligned}$$

Step 3: Verification of Deviation Equivalence: Let mixed strategies be:

$$\sigma_{t_1} = x[T_1] + (1-x)[B_1], \quad \sigma_{t_2} = y[T_2] + (1-y)[B_2], \quad \sigma_{\text{II}} = q[L] + (1-q)[R]$$

Suppose this strategy profile is not a BNE because Player 1 can profitably deviate at type t_1 by playing pure action T_1. Then:

$$U_1(\sigma \mid t_1) < U_1((T_1, \sigma_{\text{II}}) \mid t_1)$$

The ex-ante utility of Player 1 decomposes as:

$$U_1(\sigma) = p(t_1)\, U_1(\sigma \mid t_1) + p(t_2)\, U_1(\sigma \mid t_2)$$

Defining the modified strategy $\bar{\sigma}_1$ by:

$$\bar{\sigma}_1(t) = \begin{cases} T_1 & \text{if } t = t_1 \\ y[T_2] + (1-y)[B_2] & \text{if } t = t_2 \end{cases}$$

we immediately obtain $U_1(\sigma) < U_1(\bar{\sigma}_1, \sigma_{\text{II}})$, establishing that interim optimality per agent is strictly equivalent to ex-ante optimality.

2 Correlated Equilibrium

The concept of **Correlated Equilibrium**, introduced by Robert Aumann in 1974, generalizes Nash Equilibrium by allowing players to coordinate their choices through public or private signals generated by a correlation device (such as a mediator or randomizing mechanism).

2.1 Motivating Example: The Traffic Intersection (Chicken Game)

Consider two cars approaching an intersection simultaneously. Each driver can either **Cross** (C) or **Stop** (S).

	C	S
C	$(-10, -10)$	$(5, 0)$
S	$(0, 5)$	$(-1, -1)$

2.1.1 Standard Nash Equilibrium Analysis

- **Pure Strategy NE:** There are two asymmetric pure Nash equilibria:

$$(C, S) \implies \text{payoffs } (5, 0), \qquad (S, C) \implies \text{payoffs } (0, 5)$$

  Both are inherently unfair to one driver, and in a simultaneous game with no prior communication, players cannot determine which equilibrium to coordinate upon.

- **Symmetric Mixed Strategy NE:** Let Player 2 choose Cross with probability q and Stop with probability $1 - q$. For Player 1 to be indifferent:

$$U_1(C) = U_1(S) \implies -10q + 5(1 - q) = 0q - 1(1 - q)$$

$$5 - 15q = q - 1 \implies 16q = 6 \implies q = \frac{3}{8}$$

  Thus, the unique symmetric mixed strategy NE is:

$$\sigma_1^* = \sigma_2^* = \left(\frac{3}{8}[C] + \frac{5}{8}[S]\right)$$

  Under this mixed equilibrium, the expected utility of each player is:

$$U_i(\sigma^*) = \frac{3}{8}(0) + \frac{5}{8}(-1) = -\frac{5}{8} = -0.625$$

  Both players suffer negative expected payoffs, and a disastrous crash (C, C) occurs with probability $\left(\frac{3}{8}\right)^2 = \frac{9}{64} \approx 14.1\%$.

2.1.2 Introducing a Correlation Device (Traffic Light / Mediator)

Now suppose an external device (e.g., a traffic signal) selects a recommended strategy profile (s_1, s_2) according to the following joint probability distribution p:

	C	S	Total
C	0	0.55	0.55
S	0.40	0.05	0.45
Total	0.40	0.60	1.00

The recommendations are **private**: Player 1 is told only their own action s_1, and Player 2 is told only their own action s_2. Both players know the joint distribution p.

Checking Player 1's Incentive to Obey:

1. **Case 1: Player 1 is recommended Cross ($s_1 = C$):**
   - Since $p(C, C) = 0$, Player 1 updates their belief about Player 2:
   $$p(s_2 = S \mid s_1 = C) = \frac{0.55}{0 + 0.55} = 1$$
   Player 1 knows with certainty that Player 2 was told to Stop (S).
   - If Player 1 obeys and plays C: Payoff is $u_1(C, S) = 5$.
   - If Player 1 deviates to Stop (S): Payoff is $u_1(S, S) = -1$.
   - Since $5 > -1$, Player 1 has **no incentive to deviate**.
2. **Case 2: Player 1 is recommended Stop ($s_1 = S$):**
   - The marginal probability of getting recommendation S is $p(s_1 = S) = 0.40 + 0.05 = 0.45$.
   - Conditional probability that Player 2 received C:
   $$p(s_2 = C \mid s_1 = S) = \frac{0.40}{0.45} = \frac{8}{9} \approx 0.89$$
   - Conditional probability that Player 2 received S:
   $$p(s_2 = S \mid s_1 = S) = \frac{0.05}{0.45} = \frac{1}{9} \approx 0.11$$
   - If Player 1 obeys the recommendation and plays S:
   $$\mathbb{E}[u_1 \mid S] = \frac{8}{9}(0) + \frac{1}{9}(-1) = -0.11$$
   - If Player 1 disobeys and plays C:
   $$\mathbb{E}[u_1 \mid C] = \frac{8}{9}(-10) + \frac{1}{9}(5) = -\frac{80}{9} + \frac{5}{9} = -\frac{75}{9} \approx -8.33$$
   - Since $-0.11 > -8.33$, Player 1 strictly prefers to **obey the signal and stop**.

Key Takeaway

Payoff Comparison: Under this correlated distribution p, Player 1's expected payoff is:

$$U_1(p) = 0.55(5) + 0.40(0) + 0.05(-1) = 2.75 - 0.05 = 2.70$$

Notice that $2.70 \gg -\frac{5}{8} = -0.625$. Both players achieve substantially higher expected payoffs than the mixed NE, collisions are completely avoided ($p(C, C) = 0$), and yet neither player has any incentive to disobey the mediator!

2.2 Formal Definition of Correlated Equilibrium

Let $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ be a strategic form game, and let $S = \prod_{i=1}^{n} S_i$.

Definition: Extensive Form Game $\Gamma^*(p)$ of a Correlation Device

Let $p \in \Delta(S)$ be a probability distribution over the pure strategy space S. The correlation device defines an extensive form game $\Gamma^*(p)$ as follows:

1. Nature chooses a strategy profile $s = (s_1, \ldots, s_n) \in S$ according to distribution p.
2. Each player $i \in N$ is privately informed of their recommended action s_i (and learns nothing about s_{-i}).
3. Each player i then chooses an action $s_i' \in S_i$.
4. Payoffs are given by $u_i(s_1', \ldots, s_n')$.

In $\Gamma^*(p)$, a pure strategy for player i is a mapping $\tau_i : S_i \to S_i$, where $\tau_i(s_i)$ is the action taken when recommended s_i. The strategy of **following recommendations** is denoted by:

$$\tilde{\tau}_i(s_i) = s_i \quad \forall\, s_i \in S_i$$

Definition: Correlated Equilibrium (Aumann, 1974)

A probability distribution $p \in \Delta(S)$ is a **Correlated Equilibrium** of the game G if the obedience strategy profile $\tilde{\tau} = (\tilde{\tau}_i)_{i \in N}$ is a Nash Equilibrium of the game $\Gamma^*(p)$.

Key Insight: Privacy of Recommendations

It is critical that recommendations are **private**. If Player 1 knew Player 2's recommendation in advance, the strategic interaction would collapse back into standard subgames, potentially destroying the incentive to obey.

2.3 Characterization Theorem and Proof

Theorem: Linear Incentive Constraints for Correlated Equilibrium

A probability distribution $p \in \Delta(S)$ is a Correlated Equilibrium of G if and only if for every player $i \in N$ and for all actions $s_i, s_i' \in S_i$:

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(s_i, s_{-i}) \;\geq\; \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(s_i', s_{-i})$$

Proof. Fix player $i \in N$, a recommended strategy $s_i \in S_i$, and an alternative strategy $s_i' \in S_i$.

Case 1: Suppose $\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) = 0$. Then $p(s_i, s_{-i}) = 0$ for all s_{-i}. In this case, recommendation s_i occurs with probability zero, both sides of the inequality evaluate to 0, and the condition is trivially satisfied.

Case 2: Suppose $\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) > 0$. By Bayes' Rule, when player i receives recommendation s_i, their posterior belief about opponent actions is:

$$p(s_{-i} \mid s_i) = \frac{p(s_i, s_{-i})}{\sum_{t_{-i} \in S_{-i}} p(s_i, t_{-i})}$$

The expected utility of player i from following the recommendation s_i (assuming all opponents also follow their recommendations) is:

$$\mathbb{E}[u_i \mid s_i] = \sum_{s_{-i} \in S_{-i}} p(s_{-i} \mid s_i)\, u_i(s_i, s_{-i})$$

If player i instead deviates to action s_i', their expected utility is:

$$\mathbb{E}[u_i \mid s_i'] = \sum_{s_{-i} \in S_{-i}} p(s_{-i} \mid s_i)\, u_i(s_i', s_{-i})$$

By definition, $\tilde{\tau}$ is a Nash Equilibrium of $\Gamma^*(p)$ iff no player has an incentive to deviate upon receiving any recommendation s_i:

$$\sum_{s_{-i} \in S_{-i}} p(s_{-i} \mid s_i)\, u_i(s_i, s_{-i}) \;\geq\; \sum_{s_{-i} \in S_{-i}} p(s_{-i} \mid s_i)\, u_i(s_i', s_{-i}) \quad \forall\, i \in N,\ \forall\, s_i, s_i' \in S_i$$

Substituting the formula for $p(s_{-i} \mid s_i)$:

$$\frac{\sum_{s_{-i}} p(s_i, s_{-i})\, u_i(s_i, s_{-i})}{\sum_{t_{-i}} p(s_i, t_{-i})} \;\geq\; \frac{\sum_{s_{-i}} p(s_i, s_{-i})\, u_i(s_i', s_{-i})}{\sum_{t_{-i}} p(s_i, t_{-i})}$$

Multiplying both sides by the strictly positive denominator $\sum_{t_{-i}} p(s_i, t_{-i}) = p(s_i)$ yields:

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(s_i, s_{-i}) \;\geq\; \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(s_i', s_{-i})$$

which is the desired characterization. □

2.4 Properties of Correlated Equilibria

Theorem: Mixed NE Induces a Correlated Equilibrium

Every Nash Equilibrium (pure or mixed) of a strategic form game G induces a Correlated Equilibrium.

Proof. Let $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ be a Nash Equilibrium of G. Define the probability distribution p_{σ^*} on S as the product distribution:

$$p_{\sigma^*}(s_1, \ldots, s_n) = \sigma_1^*(s_1) \cdot \sigma_2^*(s_2) \cdots \sigma_n^*(s_n)$$

Since the distribution factors independently, knowing s_i provides no information about s_{-i}:

$$p(s_{-i} \mid s_i) = \sigma_{-i}^*(s_{-i})$$

The obedience condition for player i becomes:

$$\sum_{s_{-i}} \sigma_{-i}^*(s_{-i})\, u_i(s_i, s_{-i}) \;\geq\; \sum_{s_{-i}} \sigma_{-i}^*(s_{-i})\, u_i(s_i', s_{-i})$$

which is precisely the definition that s_i is a best response to σ_{-i}^* for every s_i in the support of σ_i^*. Thus, p_{σ^*} is a Correlated Equilibrium. □

Intuition

Geometric Insight: The set of Correlated Equilibria is defined by a finite system of linear inequalities together with the probability simplex constraints ($\sum_s p(s) = 1$, $p(s) \geq 0$). Therefore, **the set of Correlated Equilibria is a convex polytope**. In contrast, the set of Nash Equilibria is generally non-convex and disconnected due to the non-linear independence constraints $p(s) = \prod_i \sigma_i(s_i)$.

Example: Correlated Equilibrium in the Prisoner's Dilemma

Consider the standard Prisoner's Dilemma:

	C	D
C	$(3, 3)$	$(0, 5)$
D	$(5, 0)$	$(1, 1)$

Question: Is there any Correlated Equilibrium where the cooperative outcome (C, C) receives positive probability?

Analysis: Suppose $p(C, C) > 0$. Then Player 1 can be recommended C with positive probability. If Player 1 is recommended C, her expected payoff from obeying is:

$$\mathbb{E}[u_1 \mid C] = 3 \cdot p(C \mid C) + 0 \cdot p(D \mid C) = 3\, p(C \mid C)$$

If Player 1 deviates to D:

$$\mathbb{E}[u_1 \mid D] = 5 \cdot p(C \mid C) + 1 \cdot p(D \mid C) = 5\, p(C \mid C) + p(D \mid C)$$

Subtracting the two:

$$\mathbb{E}[u_1 \mid D] - \mathbb{E}[u_1 \mid C] = 2\, p(C \mid C) + p(D \mid C) > 0$$

Because D strictly dominates C, Player 1 has a strictly positive incentive to defect regardless of any posterior beliefs about Player 2!
Consequently, **no correlated equilibrium can assign positive probability to any strictly dominated strategy**. The unique correlated equilibrium of the Prisoner's Dilemma is $p(D, D) = 1$.

3 Introduction to Auction Theory

Auction theory models markets where goods are allocated via bidding mechanisms under asymmetric and incomplete information.

3.1 Mathematical Preliminaries: Random Variables

Definition: Cumulative Distribution Function (CDF)

For a real-valued random variable X, its **Cumulative Distribution Function (CDF)**, denoted $F_X : \mathbb{R} \to [0, 1]$, is defined by:

$$F_X(c) = \mathbb{P}(X \leq c) \quad \forall\, c \in \mathbb{R}$$

A CDF is non-decreasing, right-continuous, and satisfies $\lim_{c\to-\infty} F_X(c) = 0$ and $\lim_{c\to\infty} F_X(c) = 1$.

Definition: Probability Density Function (PDF)

If X is an absolutely continuous random variable, there exists a non-negative function $f_X : \mathbb{R} \to \mathbb{R}^+$, called the **Probability Density Function (PDF)**, such that:

$$F_X(c) = \int_{-\infty}^{c} f_X(x)\, dx \quad \forall\, c \in \mathbb{R}$$

3.2 Sealed-Bid Auctions with Independent Private Valuations (IPV)

In the **Independent Private Valuations (IPV)** model, each buyer knows their own willingness to pay for the good, but does not know the valuations of the competing bidders.

Definition: Sealed-Bid Auction Model $\langle N, (V_i, F_i)_{i\in N}, p, c\rangle$

A sealed-bid auction with independent private valuations is defined by the tuple:

$$\langle N, (V_i, F_i)_{i\in N}, p, c\rangle$$

where:

1. $N = \{1, 2, \ldots, n\}$ is the set of buyers (bidders).
2. $V_i \subseteq \mathbb{R}$ is the set of all possible valuations of buyer i. Each buyer has a private valuation $v_i \in V_i$.
3. F_i is the Cumulative Distribution Function (CDF) from which buyer i's valuation is drawn independently.
4. $p : [0, \infty)^N \to \Delta N$ is the **allocation rule**, where $p_i(b)$ denotes the probability that buyer i wins the object given the bid profile $b = (b_1, \ldots, b_n)$. Here $\Delta N = \{(p_1, \ldots, p_n) \mid p_i \geq 0,\ \sum_{i=1}^n p_i \leq 1\}$.
5. $c : N \times [0, \infty)^N \to \mathbb{R}$ (or $c : [0, \infty)^N \to \mathbb{R}^N$) is the **payment rule**, determining the payment $c_i(b)$ that buyer i must pay to the seller given the bid profile b.

3.3 Protocol of a Sealed-Bid Auction

A sealed-bid auction proceeds through the following chronological stages:

1. **Valuation Realization:** Nature draws each buyer's private valuation $v_i \in V_i$ independently according to distribution F_i.
2. **Information Revelation:** Each buyer learns their own private valuation v_i, but does not observe the valuations v_{-i} of any other buyers.
3. **Bidding:** Every buyer simultaneously submits a bid $b_i \in [0, \infty)$ in a sealed/closed envelope.
4. **Winner Determination:** The auctioneer opens all envelopes and selects the winner(s) according to the allocation rule $p(b)$. Typically, the highest bidder wins:

$$p_i(b) = \begin{cases} 1 & \text{if } b_i > \max_{j \neq i} b_j \\ \frac{1}{k} & \text{if } b_i \text{ is tied among } k \text{ highest bidders} \\ 0 & \text{otherwise} \end{cases}$$

5. **Payment Execution:** Buyers pay the seller according to payment rule $c(b)$. For example:
   - **First-Price Auction:** The winning bidder pays their own bid, $c_i(b) = b_i$.
   - **Second-Price (Vickrey) Auction:** The winning bidder pays the second-highest bid, $c_i(b) = \max_{j \neq i} b_j$.
6. **Utility Realization:** Assuming risk-neutral quasi-linear utility, buyer i's realized utility is:

$$u_i(v_i, b) = p_i(b) \cdot v_i - c_i(b)$$